Euphytica **120:** 183–194, 2001.

RFLP mapping of QTLs for photoperiod response in tropical sorghum

J. Chantereau[1,*], G. Trouche[2], J.F. Rami[3], M. Deu[1], C. Barro[4] & L. Grivet[1]
[1]*Centre de Coopération Internationale en Recherche Agronomique pour le Développement (CIRAD), TA 70/01, 34398 Montpellier Cedex 5, France;* [2]*CIRAD, 01 BP 596, Ouagadougou 01, Burkina Faso;* [3]*EURALIS GENETIQUE, 7 rue Hermès, 31520 Ramonville St-Agne, France;* [4]*Institut de l'Environnement et des Recherches Agricoles (INERA), CRREA Saria, BP10, Koudougou, Burkina Faso; (*author for correspondence)*

Received 7 March 2000; accepted 25 August 2000

Key words: maturity genes, photoperiod sensitivity, QTL, RFLP, rice, sorghum

Summary

Genetic control of flowering time in sorghum was investigated using a recombinant inbred lines population derived from a cross between IS 2807, a slightly photoperiod sensitive tropical caudatum landrace, and IS 7680, a highly photoperiod sensitive tropical guinea landrace. Progenies were sown with their parents at six different dates between 1995 and 1997 in Burkina Faso. Direct field measures and synthetic measures derived from the implementation of a model were used to characterize the photoperiod response. Emphasis was put to identify the most relevant traits to account for Basic Vegetative Phase (BVP) and photoperiod sensitivity *sensus stricto.* One QTL was detected on Linkage Group (LG) F for the traits related to BVP. Two QTLs were detected on LGs C and H for the traits related to the photoperiod sensitivity *sensus stricto*. This gives credit to at least partially independent genetic determinisms for those two components of photoperiod response. Evidences for possible orthology of the QTLs detected here with other QTLs and major genes involved in flowering time of sorghum and rice are discussed.

Introduction

Sorghum (*Sorghum bicolor* L. Moench) which was domesticated in tropical Africa, was recognised as a short day (SD) species by Garner & Allard (1923) as early as 1923. Since that time, daylength sensitivity in this species has been systematically eliminated by breeders to enlarge the range of adaptability and extend the crop area to temperate environments. Nevertheless, photoperiod sensitivity remains an important characteristic for adaptation of African landraces to the climatic resources of their environment. It is a key feature matching flowering time to length of the rainy season and securing the level and the quality of harvests (Curtis, 1968; Vaksmann et al., 1996). Consequently, a better understanding of the genetic bases will facilitate the transfer of photoperiod sensitivity from local to high yielding varieties developed for African farmers.

Several approaches have been implemented to investigate photoperiod response of sorghum both in the field (Kassam & Andrew, 1975; Hammer et al., 1989; Muchow & Carberry, 1990) and in controlled environments (Ellis et al., 1997; Alagarswamy et al., 1998). One of the less disturbing methods, because it does not use artificial lights, is based on the observation of the plant development at different sowing dates in the field in a given natural environment. To estimate the photoperiod sensitivity of a given genotype, the duration of the vegetative phase is measured for each sowing date. Then, these measures can either be used directly or they can feed a model such as the one developed by Major (1980). This author identified three genetic components to describe varietal photoperiod response: (i) the Base Vegetative Phase (BVP), defined as the shortest possible duration for floral initiation when the plants are not responsive to changes in photoperiod; (ii) the Minimum/Maximum Optimal Photoperiod (MOP), defined as the photoperiod threshold beyond which the vegetative period is influenced by changes in daylength; and (iii) the Photoperiod Sensitivity Slope (PSS), which, from the MOP, expresses the varietal linear increase to flowering time as daylength changes. The Crop Estimation through Resource and Environment Synthesis (CERES) model developed on maize by using these concepts has also been satisfactorily adapted to sorghum to characterize photoperiod sensitivity (Major et al., 1990; Alagarswamy & Ritchie, 1991; Chantereau et al., 1998; Trouche et al., 1998).

Previous mendelian genetic studies have established that four major maturity loci, designated *Ma1*, *Ma2*, *Ma3* and *Ma4*, may control flowering time in sorghum (Quinby, 1967). Furthermore, Rooney and Aydin (1999) recently showed that two other maturity loci, designated *Ma5* and *Ma6*, may also be implied in floral initiation of ultralate sorghum genotypes in the USA. Since the recent advent of molecular markers in sorghum genetics, QTL mapping for flowering time has also been documented. Based on an interspecific cross between *Sorghum bicolor* and *Sorghum propinquum*, Lin et al. (1995) identified a QTL located on linkage group (LG) D (which corresponds to LG B of Peirera et al., 1994) and assigned it to gene *Ma1*. Childs et al. (1997) mapped locus *Ma3* on LG A (LG C for Peirera et al., 1994) and concluded that this locus is a phytochrome B gene. In both cases, day-neutral and quantitative short day sorghums able to flower in temperate environments were used.

We propose here a novel investigation of the genetics of photoperiod response in sorghum based on a recombinant inbred lines (RIL) population derived from two African landraces and evaluated in the field in a tropical environment. The main objectives were: (i) to compare the ability of different traits, measured directly on plant or derived from the CERES simulation model, to account for photoperiod response; (ii) to identify genomic regions affecting the components of photoperiod response in highly photoperiod-sensitive material and compare their map location with those detected on previous studies in sorghum and rice.

Materials and methods

Plant materials

The RIL population (RIL249) consisted of 85 F_7 unselected lines derived by single seed descent from a cross between IS 2807, a poorly photoperiod sensitive caudatum local variety from Zimbabwe and IS 7680 (249 in the Cirad collection), a highly photoperiod sensitive guinea landrace from Burkina Faso (Dufour et al., 1997; Vaksmann et al., 1998). The population was developed at the Saria research station in Burkina Faso by Centre de Coopération Internationale en Recherche Agronomique pour le Développement (CIRAD, France) and Institut de l'Environnement et des Recherches Agricoles (INERA, Burkina Faso).

Field evaluation

For three years (1995–1997), the RIL lines and their progenitors were sown at different dates during the rainy season at the Saria station, Burkina Faso (2°09′W, 12°16′N, altitude 300 m). Sowing dates were 10 June 1997, 21 June 1995, 5 July 1997, 21 July 1995, 4 September 1997, 22 September 1996. At each sowing date, genotypes were arranged in a randomized complete-block design. Plots consisted of a single 5.1 m long row. Rows were spaced at 0.80 m. About one hundred grains were sown in 18 hills spaced at 0.30 m per line. Thinning to one plant was practiced at about 10 days after seedling emergence. Hills were treated with carbofuran at the sowing as a protection against insects. The plots received a standard level of fertilization (14-23-14 kg of N, P and K ha^{-1} before sowing, 22,5 kg of N ha^{-1} at beginning of stem elongation). Plots were clean weeded by hand and irrigated when necessary. Maximum and minimum air temperatures under shade were monitored daily in a meteorological shelter.

Direct measures of the photoperiod response

The traits measured in the six different experiments corresponding to the six sowing dates were indexed from 1 to 6 relatively to the julian calendar. For each sowing date and each RIL, the number of leaves and the date of flag leaf emergence at the main stem of five random plants were scored two or three times weekly from thinning. In each experiment, these data led to characterize genotypes for averaged durations of the vegetative phase from sowing to flag leaf emergence. This was established in Vegetative Calendar Time (VCT, expressed in days), in Vegetative Biological Time (VBT, expressed in number of leaves) and in Vegetative Thermal Time (VTT, expressed in degree-days, °Cd). For the thermal time, a summation of mean daily temperatures was calculated according to Alagarswamy & Ritchie (1991) with a base, optimum and maximum temperature of 8 °C, 34 °C and 44 °C, respectively.

In order to concentrate our purpose on the most informative date, we report here on the traits directly measured in experiments 1 and 5, corresponding to sowing dates June 10 and September 4, respectively. The first date corresponds to the experiment conducted under the longest day lengths, which caused the longest vegetative phases of the plants. The second date corresponds to the shortest day lengths insuring a complete vegetative growth during the rainy season, before the arrival of cold temperatures. Direct measures performed at date 6 (September 22) were not reported here because cold temperatures interacted with photoperiod and induced irregularities of the vegetative phase duration for some lines.

Photoperiod sensitivity can also be characterized by the variation of the vegetative duration between different sowing dates expressed in days (ΔVCT), leaves number (ΔVBT), and °Cd (ΔVTT), respectively. We report here the variation between the first and the fourth sowing dates (namely June 10 and July 21), that is ΔVCT_{1-4}, ΔVBT_{1-4}, and ΔVTT_{1-4}. These two dates correspond to the extremes of the linear variation of the photoperiod sensitivity (see below and Figure 1A).

The CERES sorghum model

The varietal photoperiod response was also investigated following the CERES sorghum model using the concepts of Major (1980).This model considers the period from sowing to floral initiation expressed in degree-days (Alagarswamy & Ritchie, 1991). For each RIL in each experiment, the thermal time from sowing to panicle initiation, TTINI, was obtained by multiplying by 0.6 the thermal time from sowing to flag leaf emergence, VTT. This term of 0.6 corresponds to the ratio between plastochron and phyllochron determined for sorghum (Muchow & Carberry, 1990). The day length at panicle initiation was calculated according to Forsythe et al. (1995) for a solar elevation of minus 6°, which characterize the duration of the biologically active day for photoperiod sensitive plants according to Aitken (1974).

For each RIL, TTINI for the six sowing dates was plotted against day length at estimated panicle initiation date. This permitted to estimate the three components of the photoperiod response according to Major (1980), BVP, MOP and PSS. BVP characterizes the intrinsic earliness. It was estimated as the averaged TTINI for the two September sowing dates. PSS characterizes the photoperiod sensitivity *sensus stricto*. It was estimated as the linear regression coefficient computed for the points related to the sowing dates of June and July. MOP corresponds to the daylength at the intersection of BVP and PSS. The three components are graphically illustrated on Figure 1B.

Phenotypic correlations were computed between all pairs of traits studied. A Principal Components Analysis (PCA) was also performed on the complete trait × RIL matrix in order to summarize the most striking multi-trait associations (CSTAT software; Foucher, 1992).

QTL detection

The RFLP linkage map data used for the QTL detection were those obtained by Dufour et al. (1997). This map encompassed 129 markers distributed on 12 LGs and spanning a total genetic distance of 878 cm. LGs were named according to their homology with those defined by Pereira et al. (1994). QTL analyses were performed as described by Rami et al. (1998). First, simple interval mapping method (SIM) was performed with the Plabqtl software package (Utz & Melchinger, 1995). QTL detection threshold for each trait was determined by the permutation method described by Churchill & Doerge (1994) and implemented in the Mqtl software with a global type-I error of 5% (Tinker et al., 1996). Depending on traits, LOD threshold values ranged from 2.50 to 2.75. The confidence interval for each QTL was set at a one-LOD support interval, as described by Lander & Botstein (1989). When at least one QTL was detected with SIM, composite interval mapping method (CIM) was also performed. CIM cofactors were chosen as the nearest marker of each QTL detected with SIM. CIM scans were performed using the Plabqtl software with the same LOD threshold value as in SIM. The phenotypic variance explained by a single QTL was calculated as the square of the partial correlation coefficient with the observed variable, adjusted for cofactors in case of CIM (Utz & Melchinger, 1995). The additive effect of a putative QTL was estimated by half the difference between the two homozygous classes.

Results

As expected, the distribution of the traits measuring the duration of the vegetative phases (VCT_1, VBT_1, VTT_1, VCT_5, VBT_5, VTT_5) showed that the guinea parent, IS 7680, is later maturing than the caudatum parent, IS 2807 (Figure 2). The differences between the two parents are smaller for the September sowing date. The decrease of the vegetative phase of

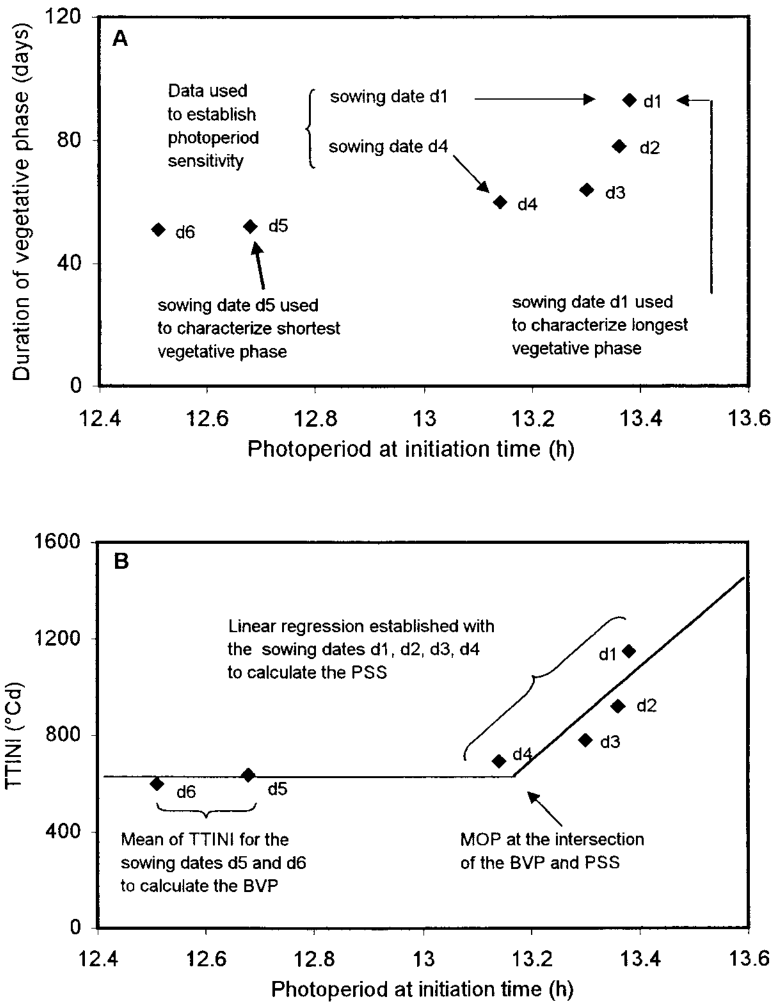

Figure 1. Traits accounting for photoperiod response and parameters of the CERES model A) Plot of sowing dates in a diagram representing the duration of the vegetative phase as a function of the photoperiod. B) Description of the components of photoperiod response according to Major (1980). From left to right, the spots corresponding to the sowing dates are in inverse order of their calendar position. Data from IS 7680, one of the two parents of the RIL249 population, were used to establish the figure.

IS 7680 regarding calendar, biological and thermal times between the first and the fourth sowing dates (ΔVCT_{1-4}, ΔVBT_{1-4}, ΔVTT_{1-4}) is larger than for IS 2807 indicating that the former is more photoperiod sensitive than the latter. Nevertheless the two parents are both photoperiod sensitive like most West African landraces (Ouattara et al., 1997).

The implementation of the CERES model permitted us to compute Major's components of the photoperiod response for all RIL. Parents are similar for MOP. They are slightly different for BVP and fairly different for PSS (Figure 2). The PSS value of IS 7680 is about twofold greater than that of IS 2807. The range of variation for all components of the photoperiod response is important in the population indicating that transgressions do occur. The values obtained for MOP are consistent with those previously reported (Major et al., 1990; Alagarswamy & Ritchie, 1991; Major & Kiniry, 1991). This is not the case for BVP and PSS for which some observed values are very large (Huda, 1987; Muchow & Carberry, 1990; Alagarswamy & Ritchie, 1991; Alagarswamy et al., 1998; Craufurd et al., 1998). This is certainly due to the tropical origin of the parents of RIL249 population which enlarge the spectrum of photoperiod response behavior compared to previous studies.

The phenotypic correlation coefficients between all pairs of traits were calculated (Table 1). VCT, VBT,

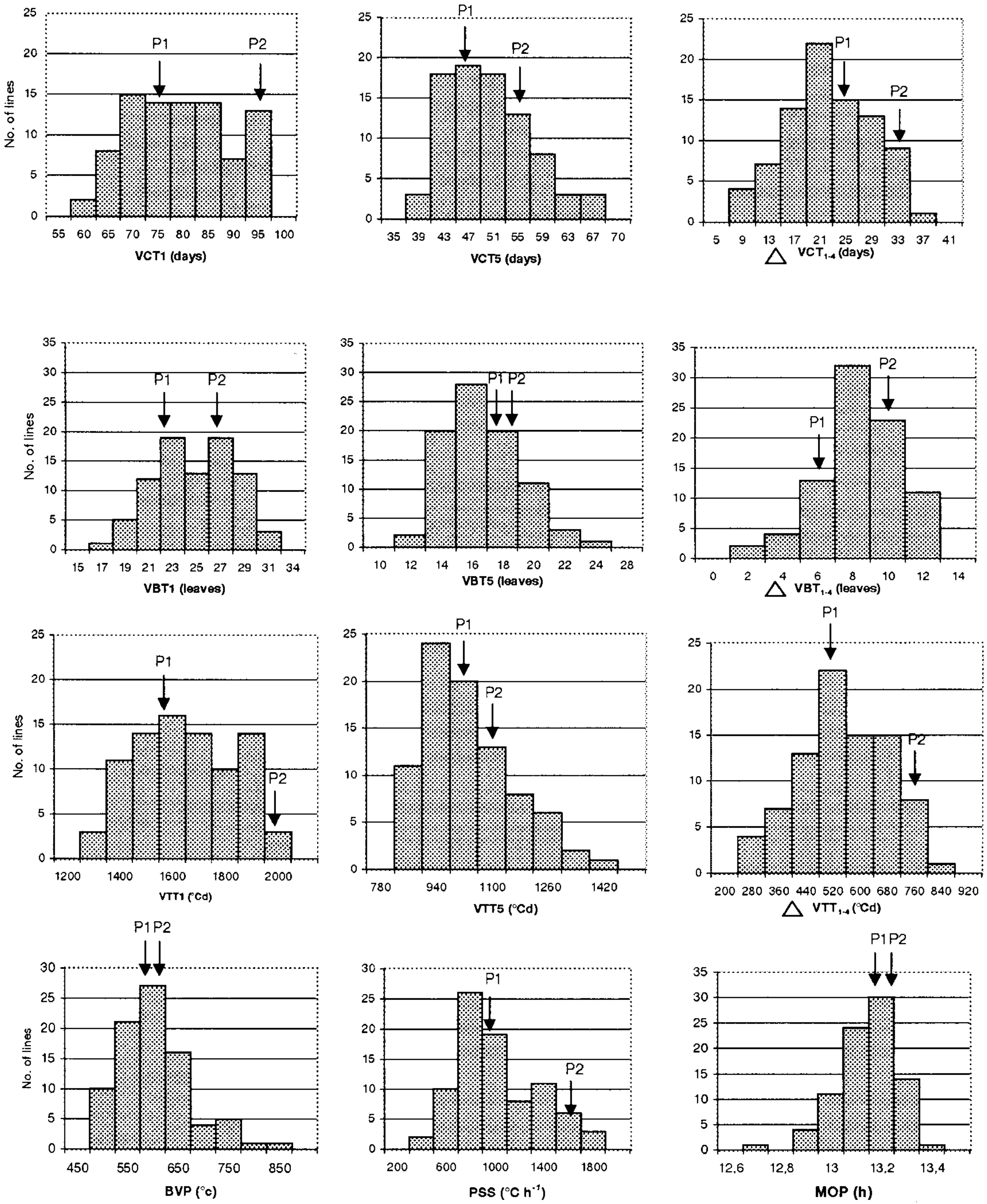

Figure 2. Frequency distribution of phenotypes for each trait for the 85 recombinant lines of the RIL249 population. Values of the two parents are shown by arrows (P1 = IS 2807, P2 = IS 7680). On the x-axis, values indicated correspond to the nearest gradation on the right.

Table 1. Phenotypic correlation coefficients between traits measured to characterize the photoperiod response in RIL249 population. Details about traits are given in the text. Correlations are given when they are significant at $p = 0.001$

	Direct measures									Ceres model		
	VCT_1	VBT_1	VTT_1	VCT_5	VBT_5	VTT_5	ΔVCT_{1-4}	ΔVBT_{1-4}	ΔVTT_{1-4}	BVP	MOP	PSS
VCT_1	1											
VBT_1	0.87	1										
VTT_1	0.99	0.87	1									
VCT_5	0.60	0.56	0.60	1								
VBT_5	0.68	0.67	0.67	0.78	1							
VTT_5	0.60	0.56	0.60	0.99	0.78	1						
ΔVCT_{1-4}	0.74	0.60	0.74	0.31	0.31	0.31	1					
ΔVBT_{1-4}	0.30	0.47	0.30	ns	ns	ns	0.59	1				
ΔVTT_{1-4}	0.77	0.62	0.77	0.33	0.33	0.33	0.99	0.59	1			
BVP	0.55	0.53	0.55	0.89	0.78	0.89	ns	ns	ns	1		
MOP	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns	1	
PSS	0.69	0.59	0.70	ns	ns	ns	0.92	0.56	0.91	ns	ns	1

VTT are always highly correlated for a given sowing date. This is due to the stability of the daily temperatures during the rainy season in the soudanian zone (Sivakumar & Gnoumou, 1987). Days are equivalent in terms of growing degree-days whatever the calendar time is. ΔVCT_{1-4} and ΔVTT_{1-4} are highly correlated (r = 0.99) but poorly correlated with ΔVBT_{1-4}. These lower correlations are due to the higher sensitivity of the biological vegetative duration, expressed in number of leaves, to errors and accidents (lower leaves can be lost prior to and during thinning). Moreover, Ellis et al. (1997) suspect that the optimum temperatures for panicle initiation and for leaf appearance differ among sorghum varieties as reported for rice. High correlations are also observed between direct measures of the photoperiod response and measures derived from the CERES model. BVP is highly correlated with all measures of the vegetative phase length performed in September (r above 0.77). PSS is highly correlated to ΔVCT_{1-4} and ΔVTT_{1-4} (r above 0.90). These results show that, in our experimental conditions, BVP and PSS can be estimated by direct relevant field measures without implementing the CERES model. No significant correlation was observed with MOP. The three components of the CERES model, BVP, MOP and PSS are not significantly correlated which suggests that they may have a distinct genetic control as already mentioned by Major (1980) and Chantereau et al. (1998).

The PCA summarized the most striking multi-trait correlations. The first two axes explained 56% and 23% of the total variation, respectively (Figure 3). Axis 1 was positively correlated with all traits but MOP. Axis 2 distinguished one group of traits related to intrinsic earliness (BVP, VCT_5, VBT_5, VTT_5) from another group of traits related to photoperiod sensitivity *sensus stricto* (PSS, ΔVCT_{1-4}, ΔVTT_{1-4}, and, at a lesser extend, ΔVBT_{1-4}). The two groups are almost independent. This summarizes correlations mentioned above. The most interesting feature brought by the PCA concerns a third group of traits, gathering the direct measures performed in June (VCT_1, VBT_1, VTT_1). This group locates at an intermediate position between the two previous groups, indicating a correlation with both of those. This highlights the likely composite nature of vegetative length measures performed in June that probably account for both BVP and PSS.

For all traits investigated, a total of eight significant QTLs were detected with SIM and 10 with CIM (Table 2). All QTLs detected with SIM were also found using CIM and CIM permitted to detect two new QTLs. QTLs were scattered over three map positions on LGs C, F, and H, respectively (Figure 4). The contribution of QTLs to phenotypic variance ranges from 13.6% to 18.0%. Two QTLs were detected on LGs F and H for both VCT1 and VTT_1 and had similar map positions. A QTL was also detected for VBT_5 at a congruent position on LG F. Two QTLs were detected for ΔVCT_{1-4} and ΔVTT_{1-4} on LGs C and H and had the same map position for both traits. On LG H, the position was also congruent with the QTLs detected for

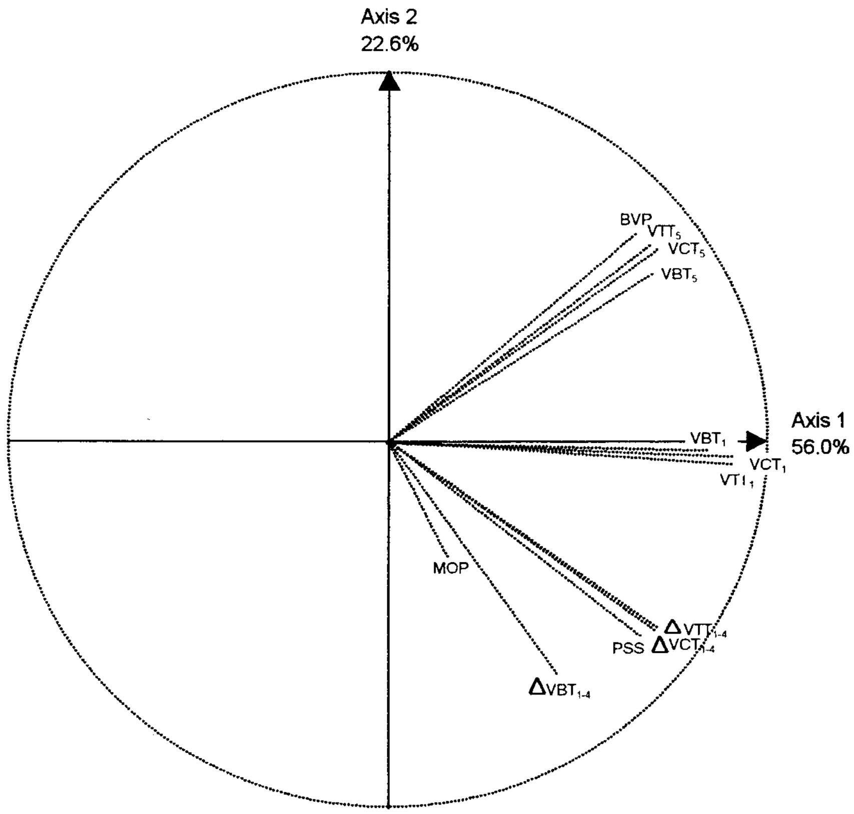

Figure 3. Projection of traits on the first plane of a Principal Components Analysis performed on the traits measuring photoperiod response in the RIL249 population.

Table 2. QTL detection results on traits related to the photoperiod response in sorghum RIL249 population. SIM is Simple Interval Mapping, CIM is Composite Interval Mapping, n is number of QTL detected, R^2 is the cumulated percentage of phenotypic variance explained by the QTLs. Details about individual QTLs are given on Figure 4

	SIM detection		CIM detection		
	N	R^2 (%)	Cofactor	n	R^2 (%)
VCT_1	1	17.6	Umc 113	2	28.8
VBT_1	0	0		0	–
VTT_1	1	17.7	Umc 113	2	28.8
VCT_5	0	–		0	–
VBT_5	1	14.2	Umc 122	1	14.2
VTT_5	0	–		0	–
ΔVCT_{1-4}	2	27.4	umc 113 – umc 37	2	27.4
ΔVBT_{1-4}	0	–		0	–
ΔVTT_{1-4}	2	26.4	umc 113 – umc 37	2	26.4
BVP	0	–		0	–
MOP	0	–		0	–
PSS	1	18.0	umc 113	1	18.0

VCT_1 and VTT_1. No QTL was found for ΔVBT_{1-4}. Traits related to the number of leaves permitted us to detect fewer QTLs than the traits related to the calendar and the thermal vegetative times. This is probably due to a lower repeatability. Regarding traits derived from the CERES model, one QTL with a major effect (18% of phenotypic variance) was detected for PSS on LG H at the same map position as QTLs identified for traits derived from direct measures. No QTL was detected for MOP and BVP. Chromosomal LOD score peak values were also reported on LGs C, F and H for traits when it exceeded 1.0 (Figure 4). Those insignificant LOD score peaks generally had map position close to significant QTLs detected for other traits. On LG F, significant and non significant LOD score peaks are found at the same map location for up to ten traits.

Discussion

In the present study we decomposed the photoperiod response into independent component by using the

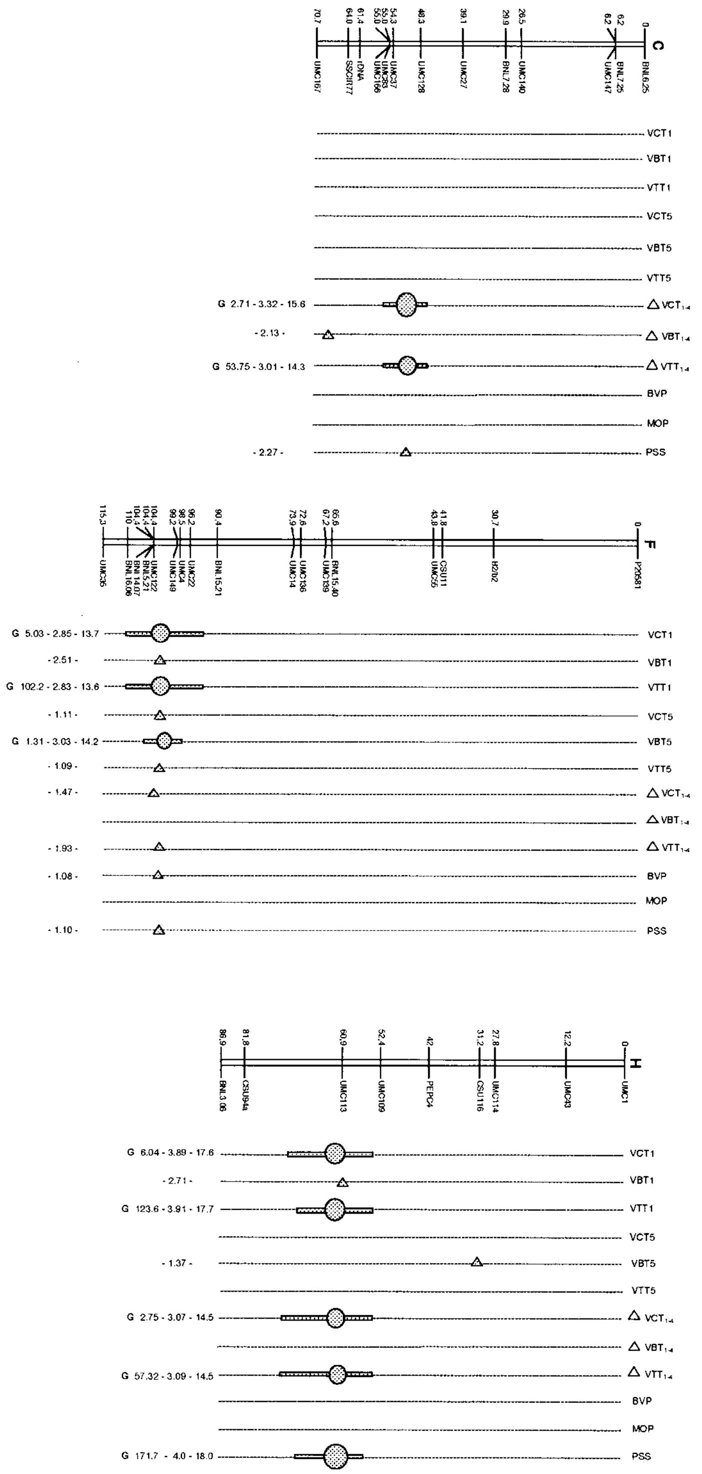

Figure 4. QTLs detected with CIM for photoperiod response in RIL249 population. Each significant QTL whose LOD score is above the significant threshold is represented with a circle at the LOD score peak position. The line represents the one-LOD score confidence interval. When a QTL is detected for at least one trait on a given LG, the LOD score peak is materialized by a triangle for the other traits when it exceed one. Values to the left of character maps are letters G or C indicating the parental origin (C for caudatum IS 2807 and G for guinea IS 7680) of the allele increasing the value of the trait, additive effect, peak LOD score value and percentage of the phenotypic variance explained by the QTL.

concept developed by Major (1980) and we tried to investigate the genetic determinism of elementary components through a QTL mapping approach. For MOP, no QTL was detected, probably due to a conjunction of a low genetic variation in our population and the difficulty to reliably measure this trait. For photoperiod sensitivity *sensus stricto* (accounted by PSS, ΔVCT_{1-4}, ΔVTT_{1-4}), QTLs were located on two map positions on LGs H and C (Figure 4). For intrinsic earliness (accounted by BVP, VCT_5, VBT_5, VTT_5), only one QTL was detected for trait VBT_5 on LG F. It is likely that the small size of the RIL249 population may have prevent to detect more QTLs, especially those with small effects. For example, Laurie (1997) already noted in wheat that the effect of genes implicated in earliness *per se* was small. These results give however a more consistent base to a distinct genetic control for intrinsic earliness and photoperiod sensitivity *sensus stricto*, which was previously suggested by Major (1980) and Chantereau et al. (1998). QTLs for traits measuring vegetative durations for the June sowing date, VCT_1 and VTT_1, were detected on LGs F and H. This can be explained by the composite nature of those two traits, which account for both, intrinsic earliness and photoperiod *sensus stricto* as it can clearly be visualized on the first plan of the PCA (Figure 3).

We have compared our results with other studies of the photoperiod response in sorghum to account for possible orthologies. Dufour (1996) detected a QTL affecting flowering date at the same position as the one detected here on LG H, in another RIL progeny derived from a cross between 379 in the Cirad collection, a guinea landrace from Senegal, and IS 2807. This QTL may be specific of tropical germplasm and could correspond to one of the two maturity genes *Ma5* and *Ma6* recently described by Rooney & Aydin (1999). Lin et al. (1995) and Paterson et al. (1995) showed that this region of LG H (I for these authors) is involved in a complex duplication with a region of LG B (D for these authors) where they located an important QTL for flowering time and a QTL for plant height (Figure 5A). They suggested that the two QTLs might respectively correspond to the *Ma1* maturity gene and the *Dw2* dwarf gene that are known to be linked (Doggett, 1988). Moreover, Pereira & Lee (1995) mapped a QTL affecting plant height on LG H in the BNL3.06 – UMC114 marker interval (that they also assumed to correspond to *Dw2*). Although colinearity is not straightforward between the two duplicated regions, this gives credit to a possible paralogy between QTLs detected on LGs H and B for both flowering time and height.

The QTLs on LG F have a close map position to the one detected by Lin et al. (1995) for days to anthesis (B in Lin et al., 1995) (Figure 5B).

The QTLs detected on LG C are located on a region where several phytochrome genes have been mapped, i.e. phytochromes A, C and B (Paterson et al., 1995), the latter being assumed by Childs et al. (1997) to be the *Ma3* gene (Figure 5C).

The comparison of map location for genes involved in the photoperiod response can also be extended to rice. The region on sorghum LG H which carries a photoperiod sensitivity QTL aligns with a region of rice chromosome 6 carrying probe RZ144 and isozyme *pgi-2* which are known to be linked to *Se-1*, a major photoperiod sensitivity gene, whereas its dominant enhancer *En-Se-1* is linked to the *wx* loci on the same chromosome (Figure 5A) (Causse et al., 1994; Kinoshita, 1995; Yano et al., 1997). Furthermore, Yano et al. (1997) identified five QTLs for heading date in rice and among them, two, *Hd-1* and *Hd-3*, might correspond to quantitative alleles of *Se-1* and *En-Se-1*, respectively. Indeed, the allelic nature of *Se-1* and *Hd-1* has been proved recently after the cloning of *Hd-1* (Yano et al., 2000). Coincidence in map position suggests that the QTL detected here on sorghum LG H might be orthologous to *Se-1* (Figure 5A).

Paterson et al. (1995) observed a colinearity between rice chromosome 7 and sorghum LG F (Figure 5B). Rice chromosome 7 is known to carry two photoperiod sensitivity genes *Se-2* and *E1* (Laurie, 1997; Yano et al., 1997; Sarma et al., 1998) that have however not been mapped precisely. Yano et al. (1997) also located *Hd-4* and*Hd-2*, two of the five QTLs previously mentioned, on chromosome 7. Moreover, they established that *Hd-4* was located nearby morphological marker *Rc* in a region collinear with that bearing QTLs for photoperiod response detected here and in the study of Lin et al. (1995) on sorghum LG F. *E1* is also known to be linked to *Rc* (Yano et al., 1997). Those observations give several tentative candidate orthologous genes in rice for the QTL detected on sorghum LG F.

Finally, sorghum LG C is partially homoeologous to rice chromosome 10 (Paterson et al., 1995) (Figure 5C). QTL implied in photoperiod sensitivity *sensus stricto* on LG C is very near probe RZ337 which is located on rice, close to the major gene *Ef-1* that controls earliness in rice (Causse et al., 1994;

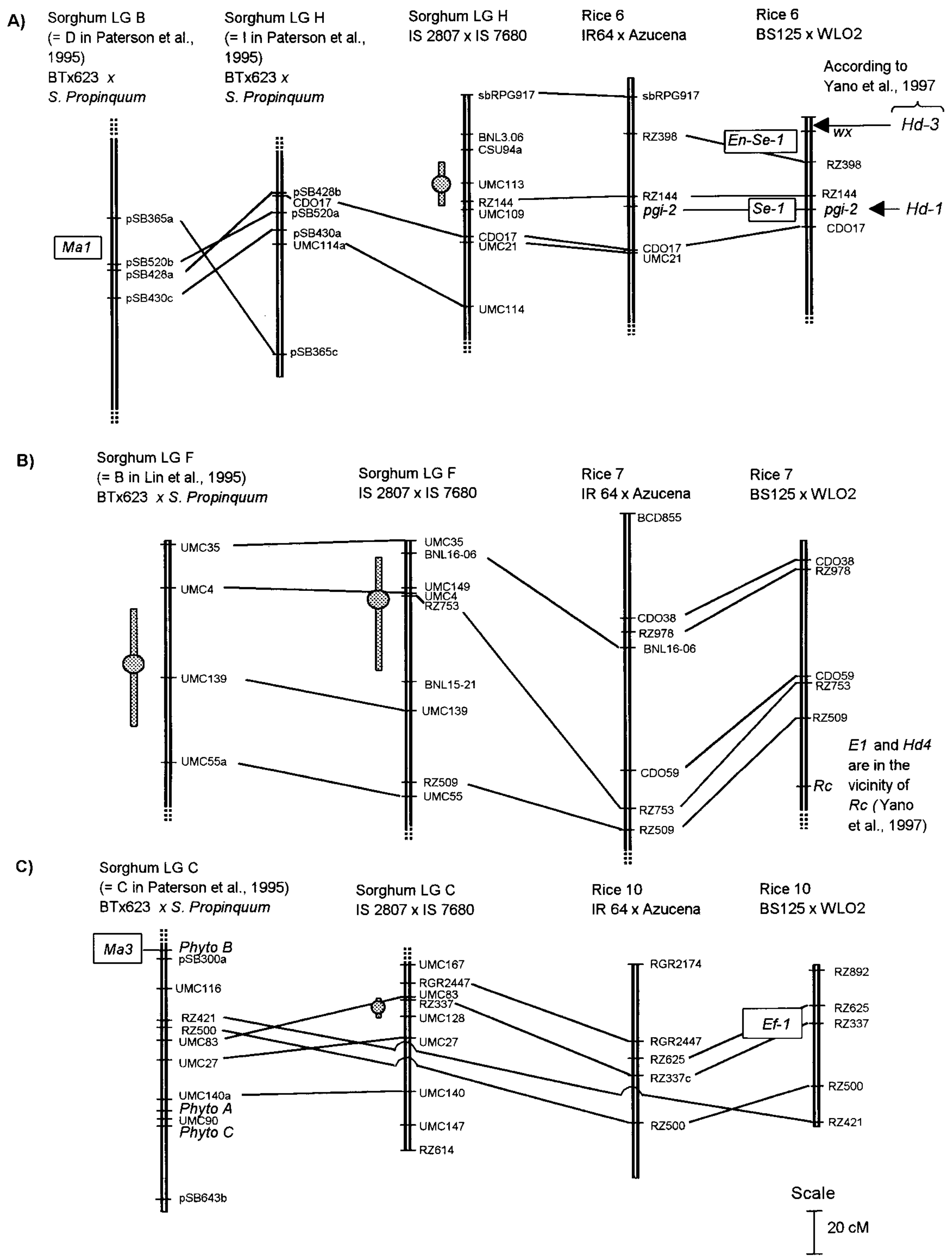

Figure 5. Comparative map position of the major genes and QTLs implied in flowering time or photoperiod sensitivity in sorghum and rice. For the map of sorghum RIL249 population derived from cross IS 2807 × IS 7680, information from Boivin et al. (1999) has been added. Information for sorghum cross BT × 623 *Sorghum bicolor* × *Sorghum propinquum* is taken from Lin et al. (1995) and Paterson et al. (1995). Information for rice cross BS 125 (*Ozyza sativa*) × WL02 (*Oryza longistaminata*) is taken from Causse et al. (1994). Some unpublished information has also been used for the map of rice cross IR64 × Azucena.

Yano et al., 1997) indicating a possible orthology between the two.

Conclusion

In this study, the photoperiod response in tropical sorghum was mostly addressed through two independent components, intrinsic earliness and photoperiod sensitivity *sensus stricto*. Simple relevant traits were identified to measure these two components and independent QTLs were detected enhancing the hypothesis of distinct genetic controls. This participates to a better understanding of the photoperiod response genetic basis, which is essential for the creation of new materials well adapted to tropical environments.

In the future, further investigations with genotypes of various origins and populations of larger size will be necessary to fully depict genetic control of photoperiod response in sorghum. Comparison with other species, especially rice, may help accelerating the process and accessing the molecular bases of the relevant loci.

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